

Our land. Our energy future.

Investigating energy resilience through electricity, seasonal resources, land constraints, industrial systems and open modelling.

THE RESEARCH QUESTION

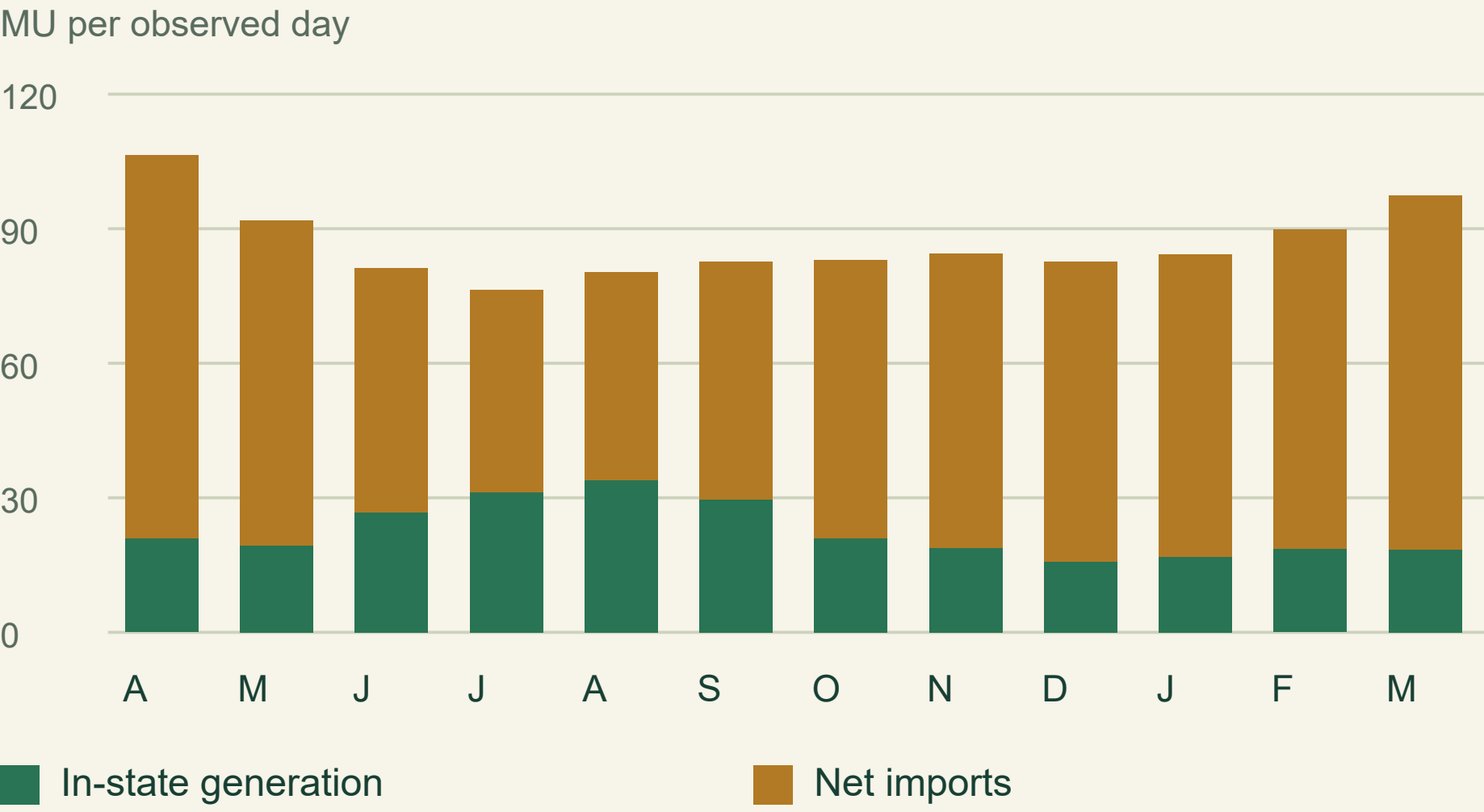
How can Kerala meet future demand reliably while respecting its land and water?

Early experiments show why timing and transfer capability matter alongside new generation. The current evidence supports comparison, not a final 2040 capacity plan.

01 / OBSERVED ELECTRICITY

A connected, seasonal system

73.8% of recorded consumption came from net imports.

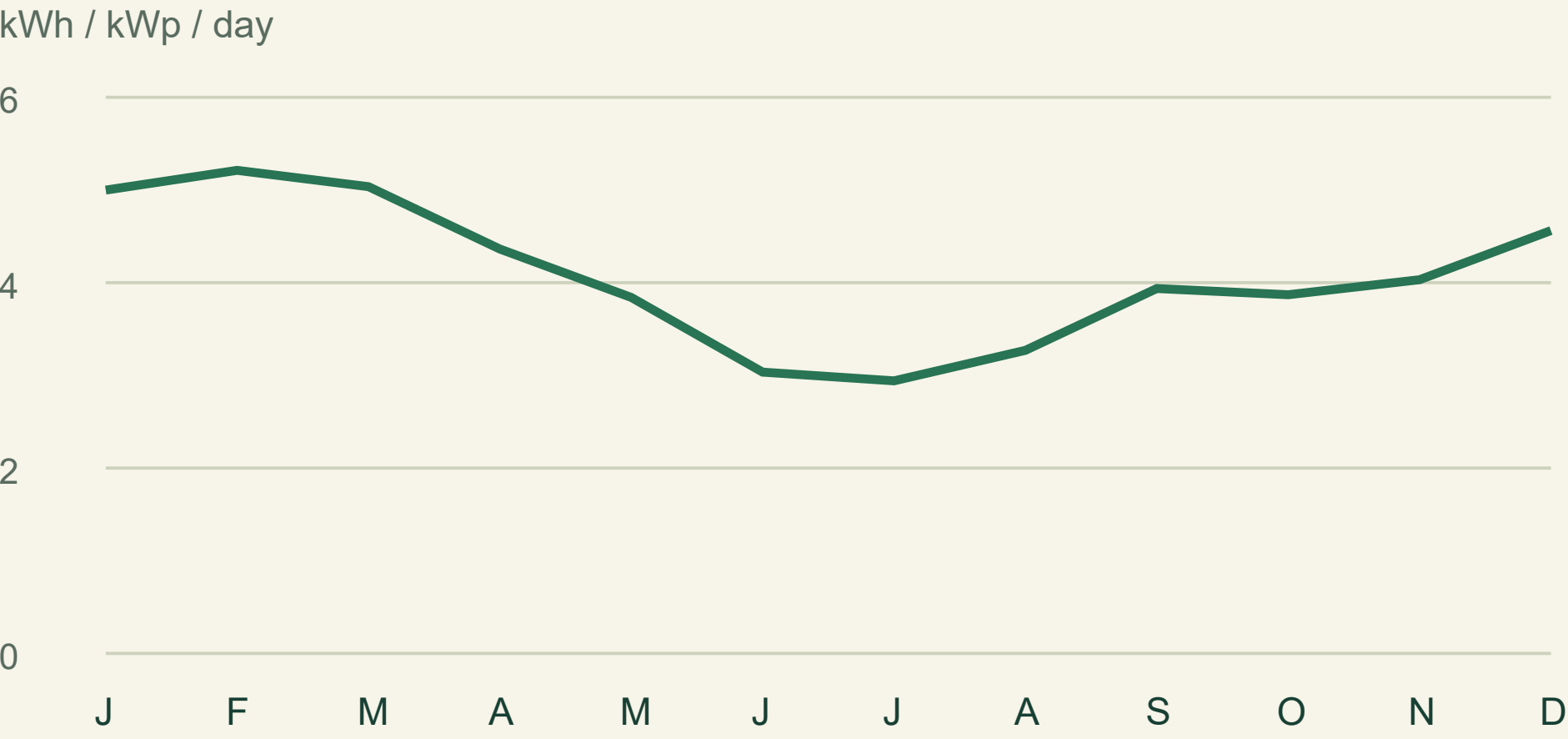


SLDC FY2024-25: 354 of 365 daily reports. The 11 missing days are not filled. Bars show available-day means, not complete monthly totals.

02 / RESOURCE & LAND

Solar follows the seasons

Monthly median source-grid PV output



Resource is not permission to build.

Global Solar Atlas 2, 1999-2018 climatology. Wind, terrain, ecology and grid access must also be checked. Eligible land and project capacity are unresolved.

03 / NEW IDUKKI RESERVOIR PILOT

Seasonal flexibility helps. Constraints remain.

12 stateful cases and 24 comparisons use the same 364-day horizon. The pilot interpolates 11 generation gaps and reconstructs a net water balance; this is not observed catchment inflow.

4,455 MW

100% transfer

0.160 GWh

unserved in the stateful pilot
1-day comparison: 11,751 GWh

3,564 MW

80% stress

367.143 GWh

unserved in the stateful pilot
1-day comparison: 1,178.137 GWh

2,673 MW

60% stress

5,131.227 GWh

unserved in the stateful pilot
1-day comparison: 5,542.266 GWh

Reference FY2030-31 demand and full Idukki availability. All comparisons use the pilot's 8,736-hour horizon, not the earlier full-year values. Numerical storage closure does not validate inflow, releases, real operations or a capacity plan.

METHOD

Preserve source records and gaps. Characterise resources. Test constrained dispatch and investment. Check numerical equivalence. Keep assumptions traceable.

WHAT STILL NEEDS WORK

Measured hourly demand; reservoir and cascade operation; outages; landed prices; statutory siting; project finance and grid feasibility.